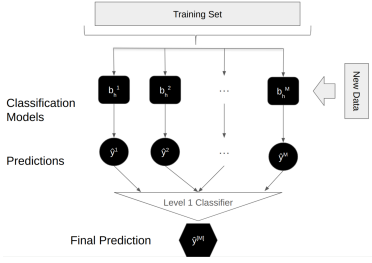
Applied Machine Learning

Ensembling: Stacking

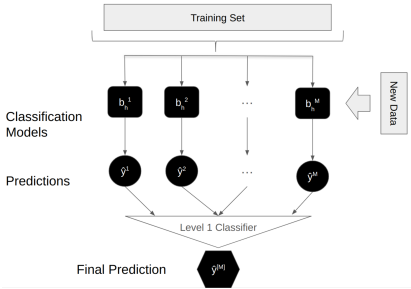


Learning goals

- What is Stacking?
- How can we stack multiple layers?

STACKING

Also called stacked generalization (▸ "Wolpert" 1992): train multiple base models (level 0) on the same data and use their predictions as input features for a level 1 model that makes the final ensemble prediction → heterogeneous ensembling



Steps:

1. Base model training (level 0)
2. Collect predictions
3. level 1 model training

In contrast to model averaging, stacking already combines predictions during training.

STACKING

Algorithm: Single Layer Stacking

Require: Training data $\mathcal{D} = \left(\left(\mathbf{x}^{(1)}, y^{(1)}\right), \ldots, \left(\mathbf{x}^{(n)}, y^{(n)}\right)\right)$

Ensure: Stacked Ensemble $f^{[M]}(\mathbf{x})$

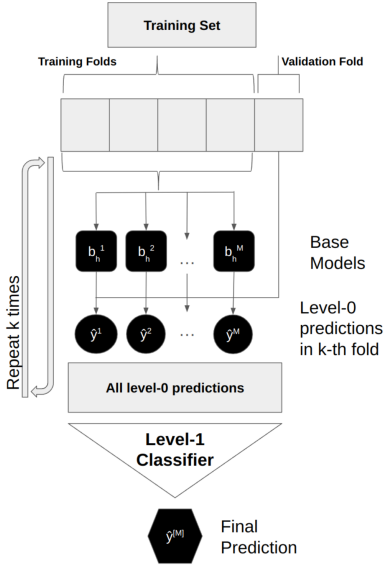
Step 1: Learn level 0 base models
for $m \leftarrow 1$ to M **do**
    Fit base model $b^{[m]}$ on $\mathcal{D}$
end for

Step 2: Construct new dataset $\mathcal{D}'$ from $\mathcal{D}$
for $i \leftarrow 1$ to n **do**
    Add $(\mathbf{x}'^{(i)}, y^{(i)})$ to new dataset, where $\mathbf{x}'^{(i)} = (b^{[1]}(\mathbf{x}^{(i)}), \ldots, b^{[M]}(\mathbf{x}^{(i)}))$
end for

Step 3: Fit level 1 model $f^{[M]}(\mathbf{x})$ on $\mathcal{D}'$
return $f^{[M]}(\mathbf{x})$

NB: We can either drop the original features or keep them when learning the level 1 model.

STACKING CROSS-VALIDATED



Using cross-validated predictions of the base models

- improve generalization
- reduce overfitting
- enhances base model diversity
- minimizes information leakage

NB: Stacking with a simple hold-out split instead of cross-validation is known as *blending*.

STACKING CROSS-VALIDATED

Algorithm: Single Layer Stacking with Cross-Validation

Require: Training data $\mathcal{D} = \left(\left(\mathbf{x}^{(1)}, y^{(1)}\right), \ldots, \left(\mathbf{x}^{(n)}, y^{(n)}\right)\right)$, number of folds K

Ensure: Stacked Ensemble $f^{[M]}(\mathbf{x})$

for $k \leftarrow 1$ to K **do**

Step 1: Learn level 0 base models

for $m \leftarrow 1$ to M **do**

Fit base model $b^{[m]}$ on $\mathcal{D}^{(k)}_{\text{train}}$

end for

Step 2: Add out-of-fold predictions to new dataset $\mathcal{D}'$

for all $(\mathbf{x}^{(i)}, y^{(i)}) \in \mathcal{D}^{(k)}_{\text{test}}$ **do**

Add $(\mathbf{x}'^{(i)}, y^{(i)})$ to new dataset, where $\mathbf{x}'^{(i)} = (b^{[1]}(\mathbf{x}^{(i)}), \ldots, b^{[M]}(\mathbf{x}^{(i)}))$

end for

end for

Step 3: Fit level 1 model $f^{[M]}(\mathbf{x})$ **on** $\mathcal{D}'$

return $f^{[M]}(\mathbf{x})$

MULTILEVEL STACKING

So far: train level 0 base models, use predictions to train the level 1 model.
Now: Also train multiple models at level 1, use predictions to train final level 2 model

1. level 0 base model training
2. Collect predictions of level 0 base models
3. level 1 model training
4. Collect predictions of level 1 models
5. level 2 final model training

NB: In principle we can repeat this training of multiple models at the l-th level indefinitely creating deep stacked ensembles.

HIGHER LEVEL MODELS

Popular choices of level 1 or level 2 models in stacking include:

- linear / logistic regression
- GES (average the predictions of the models at the previous level)

Higher-level training data:

- Probabilities instead of class labels (see ▸ "Ting and Witten" 1999)
- Include original dataset **X** or not?

FORWARD PASS / PREDICTION

Assume: trained base models $b^{[1]}, \ldots, b^{[M]}$ and meta-model $f^{[M]}(\mathbf{x})$

Require: Test data $\mathbf{X}_{\text{test}} = (\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(n_{\text{test}})})$

Ensure: Predictions $\hat{\mathbf{y}}_{\text{test}}$

Step 1: Predict with each base model

for $m \leftarrow 1$ to M **do**

$\hat{\mathbf{y}}^{[m]} \leftarrow b^{[m]}(\mathbf{X}_{\text{test}})$

end for

Step 2: Construct meta-feature matrix

$\mathbf{X}' \leftarrow [\hat{\mathbf{y}}^{[1]}, \ldots, \hat{\mathbf{y}}^{[M]}]$ or $\mathbf{X}' \leftarrow [\mathbf{X}_{\text{test}}, \hat{\mathbf{y}}^{[1]}, \ldots, \hat{\mathbf{y}}^{[M]}]$ (if original features are retained)

Step 3: Predict with the meta-model

return $f^{[M]}(\mathbf{X}')$

TWO QUESTIONS

What other methods do the following level-1 classifiers recover?

- Fit the level-1 training data with a linear model that only depends on one feature, and that feature is chosen such that ρ, our metrics of interest, is optimized.
- The level-1 model is a linear model that assign equal weight $w_m = \frac{1}{M}$ to every input feature.

AUTOGLUON

AutoGluon

https://auto.gluon.ai/

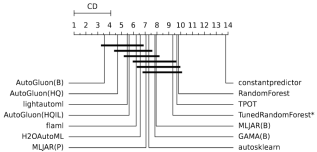


Figure: AutoGluon Performance on AMLB 1h binary classification.

- The de-facto most popular and successful AutoML system
- Developed and maintained by Amazon
- Can also work with multimodal and time-series data
- Does not use HPO to find well performing models
  - Instead, uses a set of default configurations for different learners
- Employs multi-layer stacking using the same set of default models in higher layers as for the base layer
- Uses greedy ensemble selection to combine the predictions of the last layer